

```
### Cloud Security Final Project - AWS CloudTrail Threat Analysis
## Defender Path

## objective 1: download cloudtrail logs
## step 1: setup CLI

## Install AWS vault by downloading it at
https://github.com/99designs/aws-vault

## Verify that aws CLI is installed/configured
aws --version

## configure credentials
aws configure

## AWS Access Key ID: [REDACTED]
## AWS Secret Access Key: [REDACTED]
## Default region name: us-east-1
## Default output format: json

# test that everything is configured properly
aws sts get-caller-identity
## verify that the command returns your IAM identity and account
information

## step 2: download the logs

## download the CloudTrail logs
aws s3 sync s3://flaws2-logs .
## you should get a bunch of .json.gz files within the directory path
AWSLogs...
## these logs contain CloudTrail records associated with the
simulated attack scenario
## this S3 bucket is public so that you can reference it from Athena
later

## Objective 2

## create a dedicated flaws2log directory
mkdir C:\Users\<USERNAME>\flaws2logs
cd C:\Users\<USERNAME>\flaws2logs
aws s3 sync s3://flaws2-logs .
```

```

Mode                LastWriteTime         Length Name
-----
d-----          6/12/2026   4:28 PM          flaws2logs
download: s3://flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2310Z_A1lhv3sWzzRIBFVx.json.gz to AWSLogs\653711331788\CloudTrail\us-east-1\2018\11\28\653711331788_CloudTrail_us-east-1_20181128T2310Z_A1lhv3sWzzRIBFVx.json.gz
download: s3://flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2310Z_7J9NEIxrjJsr1Xsd.json.gz to AWSLogs\653711331788\CloudTrail\us-east-1\2018\11\28\653711331788_CloudTrail_us-east-1_20181128T2310Z_7J9NEIxrjJsr1Xsd.json.gz
download: s3://flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2305Z_zK1MhON7EpHala9u.json.gz to AWSLogs\653711331788\CloudTrail\us-east-1\2018\11\28\653711331788_CloudTrail_us-east-1_20181128T2305Z_zK1MhON7EpHala9u.json.gz
download: s3://flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2310Z_rp9i9zxR2Vcpqfnz.json.gz to AWSLogs\653711331788\CloudTrail\us-east-1\2018\11\28\653711331788_CloudTrail_us-east-1_20181128T2310Z_rp9i9zxR2Vcpqfnz.json.gz
download: s3://flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2305Z_83VtWZ8Z0kiEC7Lq.json.gz to AWSLogs\653711331788\CloudTrail\us-east-1\2018\11\28\653711331788_CloudTrail_us-east-1_20181128T2305Z_83VtWZ8Z0kiEC7Lq.json.gz
download: s3://flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2310Z_jQajCuioboJDI4y.json.gz to AWSLogs\653711331788\CloudTrail\us-east-1\2018\11\28\653711331788_CloudTrail_us-east-1_20181128T2310Z_jQajCuioboJDI4y.json.gz
download: s3://flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2310Z_jW5HfNtz7KOnvcP.json.gz to AWSLogs\653711331788\CloudTrail\us-east-1\2018\11\28\653711331788_CloudTrail_us-east-1_20181128T2310Z_jW5HfNtz7KOnvcP.json.gz
download: s3://flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2235Z_cR9ra7OH1rytWXY.json.gz to AWSLogs\653711331788\CloudTrail\us-east-1\2018\11\28\653711331788_CloudTrail_us-east-1_20181128T2235Z_cR9ra7OH1rytWXY.json.gz

```

```

## verify
dir AWSLogs

```

```

Mode                LastWriteTime         Length Name
-----
d-----          6/12/2026   4:28 PM          653711331788

```

```

## verify that your AWS config file exists
notepad $HOME\.aws\config

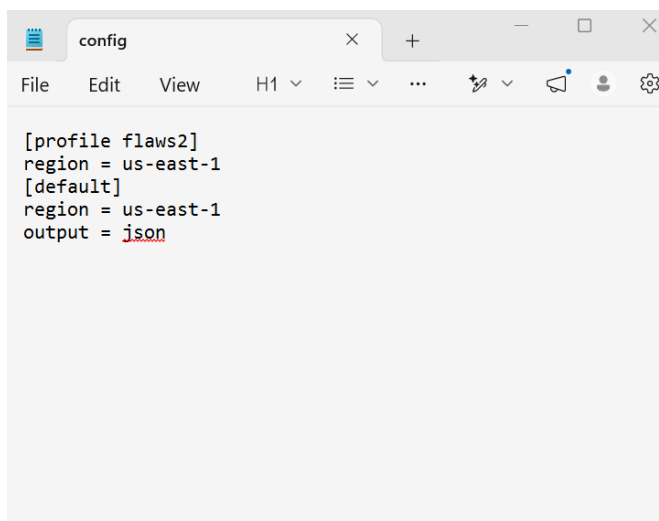
## add a profile for the target to that file:

[profile target_security]
region=us-east-1
output=json
source_profile=default
role_arn=<cross-account security role>

## save the file

## test
aws --profile target_security sts get-caller-identity

```



```
## Enumerate accessible s3 buckets
aws --profile target_security s3 ls
```

```
2018-11-20 14:50:08 flaws2.cloud
2018-11-20 13:45:26 level1.flaws2.cloud
2018-11-20 20:41:16 level2-g9785tw8478k4awxtbox9kk3c5ka8iiz.flaws2.cloud
2018-11-26 14:47:22 level3-oc6ou6dnkw8sszwvdrxrc5t5udrsw3s.flaws2.cloud
2018-11-27 15:37:27 the-end-962b72bjahfm5b4wcktm8t9z4sapemjb.flaws2.cloud
```

```
## objective 3
```

```
## install jq
winget install jqlang.jq
```

```
## re-open powershell and verify jq is installed
```

```
jq -version
```

```
## gzip isn't installed on windows but powershell can unzip .gz files
itself
```

```
Get-ChildItem -Recurse -Filter *.json.gz | ForEach-Object {
    $infile = $_.FullName
    $outfile = $infile -replace '\.gz$', ''
    $inputStream = [System.IO.File]::OpenRead($infile)
    $outputStream = [System.IO.File]::Create($outfile)
    $gzipStream = New-Object
        System.IO.Compression.GzipStream($inputStream,
        [System.IO.Compression.CompressionMode]::Decompress)
    $gzipStream.CopyTo($outputStream)
    $gzipStream.Close()
    $outputStream.Close()
    $inputStream.Close()
}
```

```
## verify the .json files exist
```

```
Get-ChildItem -Recurse -Filter *.json | Select-Object -First 5
```

Mode	LastWriteTime	Length	Name
-a----	6/12/2026 5:29 PM	3322	653711331788_ CloudTrail_us -east-1_20181 128T2235Z_cR9 ra70HlrytWyXY .json
-a----	6/12/2026 5:29 PM	1217	653711331788_ CloudTrail_us -east-1_20181 128T2305Z_83V

```
## run the jq test
Get-ChildItem -Recurse -Filter *.json | ForEach-Object {
    Get-Content $_.FullName
} | jq '.'
```

```
{
  "Records": [
    {
      "eventVersion": "1.05",
      "userIdentity": {
        "type": "AWSService",
        "invokedBy": "ecs-tasks.amazonaws.com"
      },
      "eventTime": "2018-11-28T22:31:59Z",
      "eventSource": "sts.amazonaws.com",
      "eventName": "AssumeRole",
      "awsRegion": "us-east-1",
      "sourceIPAddress": "ecs-tasks.amazonaws.com",
      "userAgent": "ecs-tasks.amazonaws.com",
      "requestParameters": {
        "roleSessionName": "d190d14a-2404-45d6-9113-4eda22d7f2c7",
        "roleArn": "arn:aws:iam::653711331788:role/Level3"
      }
    }
  ]
}
```

```
## show only event names
Get-ChildItem -Recurse -Filter *.json | ForEach-Object {
    Get-Content $_.FullName
} | jq '.Records[] | .eventName'
```

```
>> Get-Content $_.FullName
>> } | jq '.Records[] | .eventName'
'AssumeRole'
'AssumeRole'
'CreateLogStream'
'GetObject'
'GetObject'
'GetObject'
'Invoke'
'CreateLogStream'
'AssumeRole'
'CreateLogStream'
'CreateLogStream'
'ListImages'
'CreateLogStream'
'BatchGetImage'
'GetDownloadUrlForLayer'
'GetObject'
'GetObject'
'GetObject'
'GetObject'
'GetObject'
'GetObject'
'GetObject'
'Invoke'
'GetObject'
'GetObject'
'GetObject'
```

```
Windows PowerShell
tObject"
voke"
eateLogStream"
sumeRole"
eateLogStream"
eateLogStream"
stImages"
eateLogStream"
tchGetImage"
tDownloadUrlForLayer"
tObject"
tObject"
tObject"
tObject"
tObject"
tObject"
tObject"
tObject"
tObject"
voke"
tObject"
tObject"
tObject"
stObjects"
tObject"
tObject"
tObject"
tObject"
tObject"
tObject"
stBuckets"
tObject"
tObject"
```

```
## show timestamps and event names
Get-ChildItem -Recurse -Filter *.json | ForEach-Object {
    Get-Content $_.FullName
} | jq -r '.Records[] | [.eventTime, .eventName] | @tsv'
```

```

2018-11-28T22:31:59Z AssumeRole
2018-11-28T22:31:59Z AssumeRole
2018-11-28T23:03:12Z CreateLogStream
2018-11-28T23:02:56Z GetObject
2018-11-28T23:03:08Z GetObject
2018-11-28T23:03:11Z GetObject
2018-11-28T23:03:20Z Invoke
2018-11-28T23:03:50Z CreateLogStream
2018-11-28T23:03:12Z AssumeRole
2018-11-28T23:03:20Z CreateLogStream
2018-11-28T23:03:13Z CreateLogStream
2018-11-28T23:05:53Z ListImages
2018-11-28T23:03:35Z CreateLogStream
2018-11-28T23:06:17Z BatchGetImage
2018-11-28T23:06:33Z GetDownloadUrlForLayer
2018-11-28T23:02:56Z GetObject
2018-11-28T23:02:56Z GetObject
2018-11-28T23:02:56Z GetObject
2018-11-28T23:02:57Z GetObject
2018-11-28T23:03:08Z GetObject
2018-11-28T23:03:08Z GetObject
2018-11-28T23:03:08Z GetObject
2018-11-28T23:03:08Z GetObject
2018-11-28T23:03:11Z GetObject
2018-11-28T23:03:13Z Invoke
2018-11-28T23:03:14Z GetObject
2018-11-28T23:03:17Z GetObject
2018-11-28T23:03:18Z GetObject
2018-11-28T23:04:54Z ListObjects
2018-11-28T23:05:10Z GetObject
2018-11-28T23:05:12Z GetObject
2018-11-28T23:05:12Z GetObject
2018-11-28T23:07:08Z GetObject
2018-11-28T23:07:08Z GetObject
2018-11-28T23:09:28Z ListBuckets
2018-11-28T23:09:36Z GetObject

```

```

## show the useful investigation fields
Get-ChildItem -Recurse -Filter *.json | ForEach-Object {
    Get-Content $_.FullName
} | jq -r '.Records[] | [.eventName, .sourceIPAddress,
.userIdentity.arn, .userIdentity.accountId, .userIdentity.type,
.eventName] | @tsv'

## objective 4: Identify credential theft
## Review the IAM role trust policy to determine the intended
## principal and compare it against observed CloudTrail activity
aws --profile target_security iam get-role --role-name level3

```

```

{
  "Role": {
    "Path": "/",
    "RoleName": "level3",
    "RoleId": "AROAJQMBDNUMIKLZKMF64",
    "Arn": "arn:aws:iam::653711331788:role/level3",
    "CreateDate": "2018-11-23T17:55:27+00:00",
    "AssumeRolePolicyDocument": {
      "Version": "2012-10-17",
      "Statement": [
        {
          "Sid": "",
          "Effect": "Allow",
          "Principal": {
            "Service": "ecs-tasks.amazonaws.com"
          },
          "Action": "sts:AssumeRole"
        }
      ]
    },
    "Description": "Allows ECS tasks to call AWS services on your behalf.",
    "MaxSessionDuration": 3600,
    "RoleLastUsed": {
      "LastUsedDate": "2026-05-19T09:10:43+00:00",
      "Region": "us-east-1"
    }
  }
}

```

```

## objective 5: identify the public resource
## Analysis:

```

```
## The ECR repository policy contained:
## "Principal": "*"
## which allowed unauthenticated public access to several repository
actions
## this misconfiguration enabled attackers to enumerate and receive
container image information
```

```
## get the repository policy
aws --profile target_security ecr get-repository-policy
--repository-name level2
```

```
{
  "registryId": "653711331788",
  "repositoryName": "level2",
  "policyText": "{\n  \"Version\" : \"2008-10-17\",\n  \"Statement\" : [\n    {\n      \"Sid\" : \"AccessControl\",\n      \"Effect\" : \"Allow\",\n      \"Principal\" : \"*\",\n      \"Action\" : [ \"ecr:GetDownloadUrlForLayer\", \"ecr:BatchGetImage\", \"ecr:BatchCheckLayerAvailability\", \"ecr:ListImages\", \"ecr:DescribeImages\" ]\n    }\n  ]\n}"
```